

Relational Databases: Master's Level Assessment

Theoretical Foundations, Normalisation, and Cloud Deployment

COM 769 - SCALABLE ADVANCED SOFTWARE SYSTEMS

QUESTION 1: THE RELATIONAL MODEL

In E.F. Codd's foundational relational model, which of the following sets correctly maps the colloquial database terms "table," "row," and "column" to their formal academic equivalents?

A

A) Relation, Tuple, Attribute

B

B) Base Relvar, Field, Record

C

C) Entity, Tuple, Domain

D

D) Relation, Record, Element

E

E) Tuple, Relation, Attribute

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Explanation: In formal relational database theory established by E.F. Codd in the 1970s, tables are referred to as Relations (or Base Relvars), rows are referred to as Tuples (or Records), and columns are referred to as Attributes (or Fields). **Option A** provides the most accurate and standard formal mapping of these three structural elements.

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B) Base Relvar, Field, Record

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D) Relation, Record, Element

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E) Tuple, Relation, Attribute

QUESTION 2: ACID PROPERTIES

Relational database management systems typically support ACID transactions as a standard operational concern. Which of the following best defines the “Consistency” property within an ACID-compliant database?

A

A) An entire operation is fully performed or nothing is performed at all.

B

B) Any transaction will successfully bring the database from one valid state to another valid state.

C

C) Concurrent transactions will execute independently and not interfere with one another.

D

D) Once a transaction has been committed, it will remain intact even in the event of power loss.

E

E) Only structured queried elements are written to binary storage efficiently.

QUESTION 2: ACID PROPERTIES

Relational database management systems typically support ACID transactions as a standard operational concern. Which of the following best defines the “Consistency” property within an ACID-compliant database?

Explanation: ACID guarantees robust transaction processing. Atomicity ensures all-or-nothing operations; Isolation ensures concurrent operations do not interfere; and Durability ensures committed data survives system failures. Consistency specifically **dictates** that any transaction must strictly abide by all defined rules and constraints, moving the **database** from one valid state directly to another.

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B) Any transaction will successfully bring the database from one valid state to another valid state.

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C) Concurrent transactions will execute independently and not interfere with one another.

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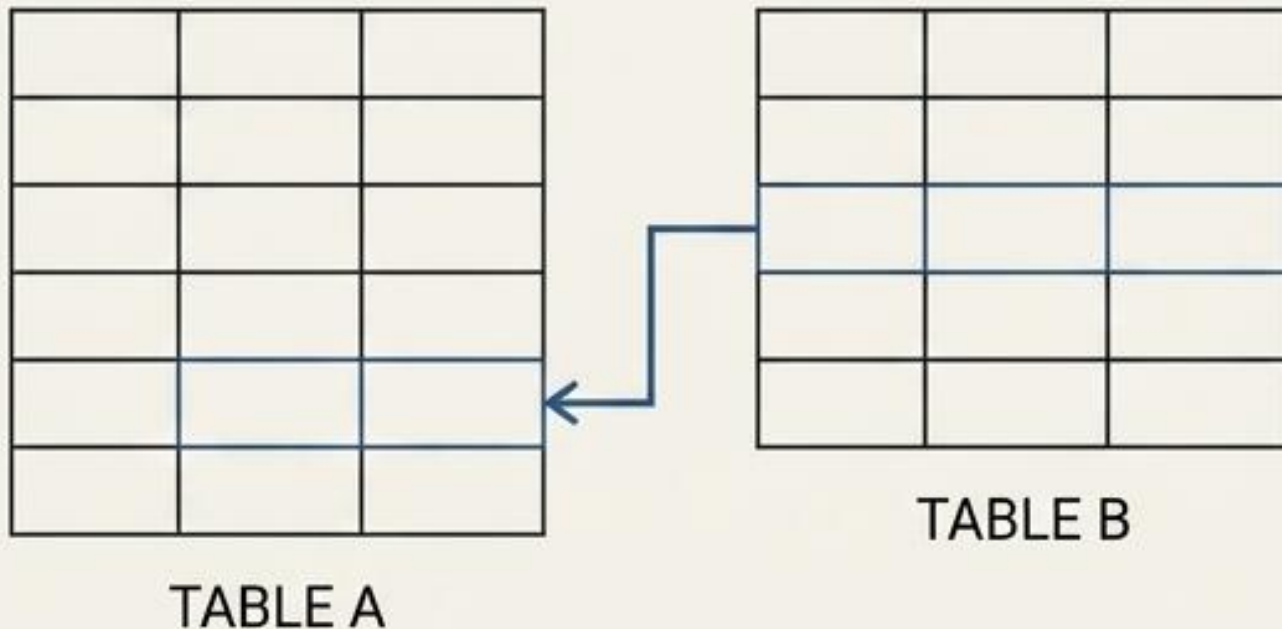
D) Once a transaction has been committed, it will remain intact even in the event of power loss.

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E) Only structured queried elements are written to binary storage efficiently.

QUESTION 3: REFERENTIAL INTEGRITY

When designing a normalised relational schema, what is the primary role of enforcing referential integrity via a Foreign Key (FK)?



- A) To allow a combination of two or more columns to act as an auto-incrementing identifier.
- B) To ensure every non-prime attribute is non-transitively dependent on the primary key.
- C) To provide assurance that related information exists by only allowing values from a specified column in a foreign table or a null value.
- D) To guarantee that a single field uniquely identifies a tuple within its own table.
- E) To prevent the storage of non-atomic textual values within an RDBMS.

QUESTION 3: REFERENTIAL INTEGRITY

When designing a normalised relational schema, what is the primary role of enforcing referential integrity via a Foreign Key (FK)?

Explanation: A Foreign Key (FK) enforces referential integrity between tables. It achieves this by dictating that the FK column can only contain values that exist in the referenced Primary Key column of the parent table (or it must be null). This prevents "orphan records" and ensures that cross-table relationships remain valid, particularly during delete or update operations.

- A) To allow a combination of two or more columns to act as an auto-incrementing identifier.
- B) To ensure every non-prime attribute is non-transitively dependent on the primary key.
- C) To provide assurance that related information exists by only allowing values from a specified column in a foreign table or a null value.
- D) To guarantee that a single field uniquely identifies a tuple within its own table.
- E) To prevent the storage of non-atomic textual values within an RDBMS.

QUESTION 4: FIRST NORMAL FORM (1NF)

A database architect is reviewing a Customer table. One of the columns, Telephone_Numbers, contains multiple phone numbers separated by commas for certain customers. To satisfy the First Normal Form (1NF), what primary action must the architect take?

A) Ensure the Telephone_Numbers attribute is fully functionally dependent on the primary key.

B) Change the data type of the column from VARCHAR to TEXT to accommodate longer strings.

C) Reorganise the data so the attribute contains only atomic values, eliminating repetition within individual records.

D) Split the table into two separate tables to resolve a transitive functional dependency.

E) Convert the primary key of the table into a composite key.

Explanation: The foundational rule of the First Normal Form (1NF) is that a table must contain only atomic (indivisible) values, and repeating groups must be eliminated. Storing a comma-separated list of telephone numbers in a single field violates this atomic requirement. The data must be restructured, typically by creating a separate table for phone numbers linked by a foreign key.

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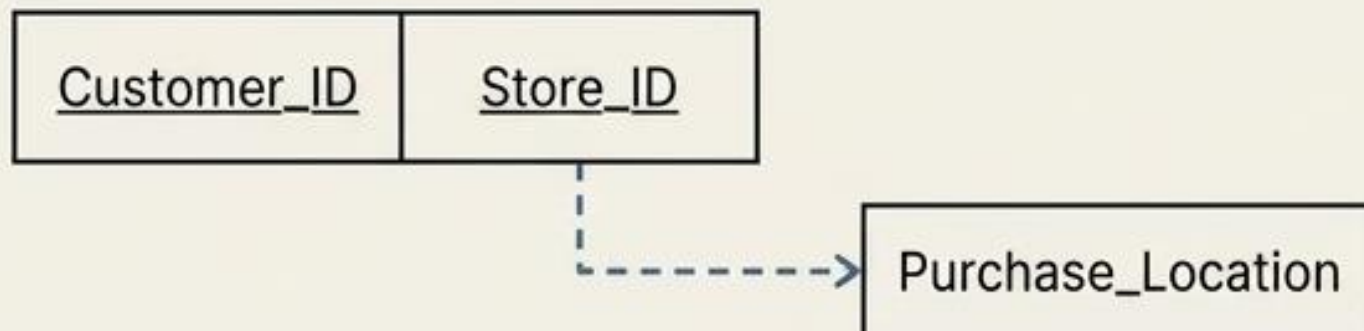
C) Reorganise the data so the attribute contains only atomic values, eliminating repetition within individual records.

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QUESTION 5: SECOND NORMAL FORM (2NF)

A relation capturing store purchases uses a Composite Key (CK) consisting of Customer_ID and Store_ID. The table includes an attribute called Purchase_Location. If Purchase_Location is dependent only on Store_ID, why does this violate the Second Normal Form (2NF)?



A) The table lacks an auto-incrementing Primary Key.

B) A non-key attribute is not fully functionally dependent on the primary key, depending only on a partial element of the composite key.

C) The Purchase_Location attribute exhibits a transitive dependency on another non-prime attribute.

D) The combination of Customer_ID and Store_ID fails to uniquely identify a tuple.

E) The table contains non-atomic variables.

Explanation: 2NF requires that a table first satisfies 1NF, and additionally, that all non-key attributes are fully functionally dependent on the entire primary key. In this scenario, Purchase_Location depends only on Store_ID (half of the composite key), not on the combination of Customer_ID and Store_ID. This partial dependency violates 2NF.

A) The table lacks an auto-incrementing Primary Key.

B) A non-key attribute is not fully functionally dependent on the primary key, depending only on a partial element of the composite key.

C) The Purchase_Location attribute exhibits a transitive dependency on another non-prime attribute.

D) The combination of Customer_ID and Store_ID fails to uniquely identify a tuple.

E) The table contains non-atomic variables.

QUESTION 6: THIRD NORMAL FORM (3NF)

Bill Kent famously summarised E.F. Codd's Third Normal Form (3NF) as:

“Every non-key attribute must provide a fact about the key, the whole key, and nothing but the key.”

Which of the following defines a violation of 3NF?

A) A table allows a non-prime attribute to be transitively dependent on the primary key via another non-prime attribute.

B) A table utilises a composite key instead of a unique numerical identifier.

C) A non-key attribute is partially dependent on a composite key element.

D) A table contains repeating groups of data in a single row.

E) A foreign key permits null values.

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"Every non-key attribute must provide a fact about the key, the whole key, and nothing but the key."

Which of the following defines a violation of 3NF?

Explanation: 3NF requires that a database satisfy 2NF and that every non-prime attribute is non-transitively dependent on every key. For example, if Book_ID determines Genre_ID, and Genre_ID determines Genre_Type, then Genre_Type is transitively dependent on Book_ID. This must be resolved by splitting the data into separate tables to achieve 3NF.

A) A table allows a non-prime attribute to be transitively dependent on the primary key via another non-prime attribute.

B) A table utilises a composite key instead of a unique numerical identifier.

C) A non-key attribute is partially dependent on a composite key element.

D) A table contains repeating groups of data in a single row.

E) A foreign key permits null values.

QUESTION 7: ER-MAPPING AND CARDINALITY

When designing an Entity-Relationship schema for a library application, you must define the relationship between a Books table and an Authors table. Given that a specific book may be co-authored by several writers, and an individual writer may author multiple different books, which mapping form is required?

A) One-to-One (1:1)

B) One-to-Many (1:M)

C) Many-to-One (M:1)

D) Many-to-Many (M:M)

E) Recursive

Explanation:

When a single record in Table A can relate to multiple records in Table B, and a single record in Table B can relate to multiple records in Table A, the relationship is Many-to-Many (M:M).

In relational databases, this is typically resolved by creating a third "junction" or "linking" table that holds foreign keys referencing the primary keys of both original tables.

A) One-to-One (1:1)

B) One-to-Many (1:M)

C) Many-to-One (M:1)

D) Many-to-Many (M:M)

E) Recursive

QUESTION 8: SQL LANGUAGE ELEMENTS

Structured Query Language (SQL) is subdivided into multiple distinct language elements. Conceptually, how does a SQL “Predicate” differ from a SQL “Statement”?

A) A predicate evaluates a mathematical function, while a statement retrieves data.

B) A predicate combines rows from multiple tables, while a statement restricts result sets.

C) A predicate is an instruction that alters the database schema, while a statement is the result set itself.

D) A predicate is a condition to be evaluated (such as within a WHERE clause), whereas a statement is an instruction that may have an effect on data or database schema.

E) There is no functional difference; the terms are strictly interchangeable in modern RDBMS.

Explanation:

In SQL taxonomy, a statement is the overarching instruction sent to the database to perform an action (e.g., `UPDATE country SET pop. + 1 WHERE name = 'USA';`). Within that statement, a predicate is the specific condition that evaluates to true, false, or unknown (e.g., `name = 'USA'`), which filters the data the statement acts upon.

A) A predicate is a statement to be evaluated (such as within a WHERE clause), whereas a statement is an instruction that may have an effect on data or database schema.

B) A predicate is a statement to be evaluated (such as within a WHERE clause), whereas a statement is an instruction that may have an effect on data or database schema.

C) A predicate is a condition to be evaluated (such as within a WHERE clause), whereas a statement is an instruction that may have an effect on data or database schema.

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QUESTION 9: SQL QUERY CLAUSES

A data analyst needs to write a SELECT query that collects individual transaction records, consolidates them by regional branches, and then presents the final output sorted chronologically. Which two SQL clauses are essential for achieving this specific aggregation and sorting?

A) JOIN and DISTINCT

B) GROUP BY and ORDER BY

C) WHERE and LIMIT

D) INSERT and UPDATE

E) DISTINCT and ORDER BY

Explanation:

The GROUP BY clause is used in a SELECT statement to collect data across multiple records and group the results by one or more columns (e.g., consolidating by consolidating by regional branches). The ORDER BY clause is then used to sort the resulting grouped data set by a specified specified column list (e.g., sorting chronologically).

A) JOIN and DISTINCT

B) GROUP BY and ORDER BY

C) WHERE and LIMIT

D) INSERT and UPDATE

E) DISTINCT and ORDER BY

QUESTION 10: RDBMS DATA TYPES

Database performance is heavily reliant on correct data type selection. Which of the following accurately describes a critical functional limitation of the TEXT data type in most SQL implementations?

A) It can only store strings up to a maximum of 255 characters.

B) It fundamentally stores binary values rather than alphanumeric characters.

C) Strings stored as TEXT generally cannot be ordered or used within a WHERE clause.

D) It forces the database engine to bypass ACID compliance for that specific table.

E) It is strictly reserved for dynamically sized JSON or XML arrays.

Explanation:

While data types like CHAR and VARCHAR are designed for indexed string manipulation and filtering, the TEXT data type is intended for storing large bodies of unstructured string data. Because of the way TEXT blobs are stored and indexed, they generally cannot be used efficiently in ORDER BY clauses or evaluated directly in WHERE clause predicates in most RDBMS environments.

A) JOIN and DISTINCT

B) GROUP BY and ORDER BY

C) Strings stored as TEXT generally cannot be ordered or used within a WHERE clause.

D) INSERT and UPDATE

E) DISTINCT and ORDER BY